S. Hun Baek

Atmospheric, Earth and Energy Division
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Professional Experience

***Staff Scientist*, Lawrence Livermore National Laboratory**	**8/2026-present**
Postdoctoral Researcher, Lawrence Livermore National Laboratory	9/2023-8/2026
Climate Scientist – Physical Hazard Analysis, S&P Global Sustainable1	9/2022-6/2023
Postdoctoral Associate, Yale University (50% appointment 9/22-6/23)	9/2020-8/2023

Education

Ph.D., Earth & Environmental Sciences, Columbia University	2020
B.A., Environmental Science, Columbia University	2014

Fellowships and Awards

- Stand Out Award, LLNL Atmospheric, Earth, & Energy Division (2026)
- Richard Foster Flint Postdoctoral Fellowship, Yale University (2020-2022)
- 1st Place ($500), Expert Scored Poster Presentation, Yale Climate Day (2021)
- Nominee, Schmidt Science Fellows (2019)
- Honorable Mention, National Science Foundation Research Fellowship (2016)
- Dean's Fellow (5-yr full funding), Columbia University (2015-2020)

Funding and Grants

2027-2030	DOE Office of Science, HyperFORMANCE: A Framework for Optimization, Research, Modeling and AI with Nonlinear Coupled Energy-Land-Water Systems (senior personnel/co-investigator)
2024-2027	DOE Earthshots, Carbon Dioxide Removal and High-Performance Computing: Planetary Boundaries of Earth Shots ($5.7 million; co-PI and institutional PI)
2020-2022	Richard Foster Flint Postdoctoral Fellowship of Yale University ($130K awarded on original proposal; extended one year)

Invited Talks

2026 *Downscaling Workstream Teleconference*, Lawrence Livermore National Laboratory
2026 *Seminar*, American Museum of Natural History
2026 *Research Applications Laboratory Seminar*, National Center for Atmospheric Research
2025 *Physical and Life Sciences Post Doc Seminar*, Lawrence Livermore National Laboratory
2025 *Downscaling Workstream Teleconference*, Lawrence Livermore National Laboratory
2025 *Environmental Sciences Seminar*, University of Virginia
2024 *Environmental Studies and Sciences Seminar*, Santa Clara University
2023 *Current Research in Earth, Environmental and Planetary Sciences*, Rice University
2023 *Seminar,* Department of Earth and Environmental Sciences, University of Texas-Arlington
2022 *Seminar,* Department of Earth and Environmental Sciences, University of Illinois Chicago
2021 *Atmosphere Ocean Climate Dynamics Seminar*, Yale University
2021 *Departmental Seminar,* Pohang University of Science and Technology
2020 *Informal Seminar,* Geophysical Fluid Dynamics Laboratory

Peer Reviewed Publications

24. **S.H. Baek**, C.B. Skinner, P.A. Ullrich, J.M. Lora, P.A. Raymond, J.E. Smerdon. Reorganization of atmospheric moisture sources and precipitation recycling over North America Under Future Warming, *in prep.*
23. C.B. Skinner, D. Swain, D. Singh, B. Mathew, **S.H. Baek**. Observed and Projected Lengthening of the Extreme Heat Season, *submitted*.
22. ARTMIP Consortium (including **S.H. Baek**; 202x). The ARTMIP Tier 2 Polar Synthesis: A comparative analysis of polar atmospheric river tracking methods in historical and future climate states. *The Cryosphere*, *submitted*.
21. **S.H. Baek** and P.A. Ullrich. Nominal grid spacing in high-resolution precipitation products obscures effective resolution and watershed-scale extremes, *submitted.*
20. ARTMIP Consortium (including **S.H. Baek**; 202x). Lessons from ten years of the atmospheric river tracking method intercomparison project. *Bulletin of the American Meteorological Society*, *in review*.
19. **S.H. Baek**, C.M. Zarzycki, P.A. Ullrich, C. Randall, A.K. Srivastava, W.Y. Wu, R.R. McCrary, A. Rhoades (2026). Weather sensitivity to sea surface temperatures over the contiguous US in a medium ensemble of high-resolution regionally refined simulation using E3SM, *Journal of Advances in Modeling Earth Systems*, https://doi.org/ 10.1029/2025MS005591
18. W.D. Rush and **S.H. Baek** (2026). Solar Variability Impacts on Global and Regional Climate: A Comprehensive Review. Encyclopedia of Climate System Science (Major Reference Work)
17. **S.H. Baek**, J.M. Lora, C.B. Skinner, M. Fu, J. Zhu (2025). Atmospheric and oceanic energetics during North Atlantic freshening events, *Climate Dynamics*, https://doi.org/10.1007/s00382-025-07761-1.
16. **S.H. Baek**, E.E. McClenny, P.A. Ullrich (2025). Response of atmospheric river width and intensity to aquaplanet warming: A detection algorithm- and background moisture-independent approach, *Journal of Geophysical Research – Atmospheres*, https://doi.org/10.1029/2025JD043367.
15. **S.H. Baek**, P.A. Ullrich, B. Dong, J. Lee (2024). Evaluating downscaling with expected hydroclimatic co-variances, *Geoscientific Model Development*, https://doi.org/10.5194/gmd-17-8665-2024.
14. J.M. Lora, C.B. Skinner, W.D. Rush, **S.H. Baek** (2023). The hydrologic cycle and atmospheric rivers in CESM2 simulations of the Last Glacial Maximum, *Geophysical Research Letters,* https://doi.org/10.1029/2023GL104805
13. **S.H. Baek***, Y. Kanzaki*, J.M. Lora, N. Planavksy, C. Reinhard, S. Zhang (2023). Impact of climate on the global capacity for enhanced rock weathering on croplands, *Earth's Future*, https://doi.org/10.1029/2023EF003698 (*denotes equal contribution).
12. **S.H. Baek**, J.M. Battalio, J.M. Lora (2023). Atmospheric river variability over the last millennium driven by annular modes, *AGU Advances*, https://doi.org/10.1029/2022AV000834.
11. **S.H. Baek**, Y. Kushnir, M. Ting, J.E. Smerdon, J.M. Lora (2022). Regional Signatures of Forced North Atlantic SST Variability: A Limited Role for Aerosols and Greenhouse Gases, *Geophysical Research Letters,* https://doi.org/10.1029/2022GL097794.
10. Scholz, S.R., R. Seager, M. Ting, Y. Kushnir, J.E. Smerdon, B.I. Cook, E.R. Cook, **S.H. Baek** (2022). Changing hydroclimate dynamics and the 19th to 20th century wetting trend in the English Channel region of northwest Europe, *Climate Dynamics,* https://doi.org/10.1007/s00382-021-05977-5.

Peer Reviewed Publications (continued)

9. **S.H. Baek**, Y. Kushnir, W.A. Robinson, J.M. Lora, D.E. Lee, M. Ting (2021). An Atmospheric Bridge Between the Subpolar and Tropical Atlantic Regions: A Perplexing Asymmetric Teleconnection, *Geophysical Research Letters*, https://doi.org/10.1029/2021GL096602
8. **S.H. Baek** and J.M. Lora (2021). Counterbalancing influences of aerosols and greenhouse gases on atmospheric rivers, *Nature Climate Change,* https://doi.org/10.1038/s41558-021-01166-8.
7. **Baek, S.H.**, J.E. Smerdon, B.I. Cook, A.P. Williams (2021). US Pacific Coastal Droughts are Predominantly Driven by Internal Atmospheric Variability, *Journal of Climate*, https://doi.org/10.1175/JCLI-D-20-0365.1
6. **Baek, S.H.**, J.E. Smerdon, G. Dobrin, J.G. Naimark, E.R. Cook, B.I. Cook, R. Seager, M.A. Cane (2020). A quantitative hydroclimatic context for the European Great Famine of 1315-1317, *Communications Earth & Environment,* https://doi.org/10.1038/s43247-020-00016-3
5. Williams, A.P., E.R. Cook, J.E. Smerdon, B.I. Cook, J.T. Abatzoglou, K. Bolles, **S.H. Baek**, A. Badger, B. Livneh (2020). Large contribution from anthropogenic warming to a developing North American megadrought, *Science*, https://doi.org/10.1126/science.aaz9600
4. **Baek, S.H.**, N.J. Steiger, J.E. Smerdon, R. Seager (2019). Oceanic drivers of spatially widespread droughts in the contiguous US over the Common Era, *Geophysical Research Letters*, https://doi.org/10.1029/2019GL082838
3. **Baek, S.H.**, J.E. Smerdon, R. Seager, A. P. Williams, B. I. Cook (2019). Pacific Ocean forcing and atmospheric variability are the dominant causes of spatially widespread droughts in the contiguous United States, *Journal of Geophysical Research – Atmospheres,* https://doi.org/10.1029/2018JD029219.
2. PAGES Hydro2k Consortium (including **S.H. Baek**; 2017), Comparing proxy and model estimates of hydroclimate variability and change over the Common Era, *Climate of the Past*, https://doi.org/10.5194/cp-13-1851-2017.
1. **Baek, S.H.**, J.E. Smerdon, S. Coats, A. P. Williams, B.I. Cook, E.R. Cook, and R. Seager (2017), Precipitation, temperature, and teleconnection signals across the combined North American, Monsoon Asia, and Old World Drought Atlases, *Journal of Climate*, https://doi.org/10.1175/JCLI-D-16-0766.1

Select Conference Presentations

2025 **Baek, S.H.,** E.E. McClenny, P.A. Ullrich, K.T. Quagraine, W.D. Collins. Response to atmospheric river width and intensity to aquaplanet warming: A detection algorithm- and background moisture-independent approach. (Oral). AGU Atmospheric Rivers: Processes, Impacts, Observations, and Uncertainties

2024 **Baek, S.H.**, P.A. Ullrich, B. Dong, J. Lee. Evaluating downscaling with expected hydroclimatic co-variances. (Oral). AGU Climate Downscaling and Weather Postprocessing: Development, Evaluation, and Applications II.

2023 **Baek, S.H.**, Y. Kanzaki, J.M. Lora, N. Planavksy, C. Reinhard, S. Zhang. Impact of climate on the global capacity for enhanced rock weathering on croplands (Poster). *AGU Fall Meeting 2023: Science-Based Gigaton-Scale Carbon Dioxide Removal (CDR): Strategies for Monitoring, Measurements, Reporting, and Verification (MMRV) II*

2022 **Baek, S.H.**, J.M. Battalio, J.M. Lora. Atmospheric river variability over the last millennium driven by annular modes (Oral). *AGU Fall Meeting 2022: Atmospheric Rivers: Processes, Impacts, and Uncertainties*

Select Conference Presentations (continued)

2022 **Baek, S.H.**, J.M. Battalio, J.M. Lora. Atmospheric river variability over the last millennium driven by annular modes (Oral). *4th International Atmospheric River Conference 2022*

2022 **Baek, S.H.**, Y. Kushnir, W.A. Robinson, J.M. Lora, D.E. Lee, M. Ting. An Atmospheric Bridge Between the Subpolar and Tropical Atlantic Regions: A Perplexing Asymmetric Teleconnection. (Oral). *EGU General Assembly 2022: The North Atlantic: Natural Variability and Global Change*.

2022 **Baek, S.H.**, Y. Kushnir, W.A. Robinson, J.M. Lora, D.E. Lee, M. Ting. An Atmospheric Bridge Between the Subpolar and Tropical Atlantic Regions: A Perplexing Asymmetric Teleconnection. (Oral). *US AMOC Science Team Meeting*.

2022 **Baek, S.H.** and J.M. Lora. Counterbalancing Influences of Aerosols and GHGs on Atmospheric Rivers. (Oral). *American Meteorological Society: 35th Conference on Climate Variability and Change*.

2021 **Baek, S.H.** and J.M. Lora. Counterbalancing Influences of Aerosols and GHGs on Atmospheric Rivers. (Poster). *Yale Climate Day*.

2018 **Baek, S.H.**, N.J. Steiger, J.E. Smerdon. Parsing the dominant ocean influences on spatially widespread droughts in the contiguous US over the Common Era (Oral). *AGU Fall Meeting 2018: Climate of the Common Era*.

2017 **Baek, S.H.**, J.E. Smerdon, S. Coats, A.P. Williams, B.I. Cook, E.R. Cook, R. Seager. Precipitation, temperature, and teleconnection signals across the combined North American, Monsoon Asia, and Old World Drought Atlases (Poster). *PAGES OSM Open Science Meeting: Large-scale hydroclimate variability and change of the Common Era: Patterns, Impacts, and Processes*.

Community Service

- *Organizer*, LLNL Climate and Weather Seminar Series (10/2025 - 10/2026)
- *Co-convener*, Decision-relevant understanding of impactful weather and extremes session, AGU 2026.
- *Co-convener*, Decision-relevant understanding of impactful weather and extremes session, AGU 2025.
- *Co-convener*, The atmospheric water cycle: processes, dynamics, isotopic tracers, and characteristics session, EGU 2024.
- *Volunteer*, LLNL Recruiting Booth, AGU 2023
- *Volunteer*, Do-It-Yourself-Dynamics (*DIYnamics*), 2023
- *Volunteer*, Lamont Open House, 2016
- *Co-experiment Leader*, Girls Science Day, 2015

Teaching Experience

2024 Spring *Guest Lecturer*, ECS0003 Introduction to Climate Science
2019 Fall *Teaching Assistant,* UN2200 Earth Environmental System: Earth System
2017 Spring *Teaching Assistant*, UN2310 Earth Environmental System: Life System
2016 Fall *Teaching Assistant,* W4008 Introduction to Atmospheric Science

Peer Review Service

- *National Science Foundation (proposal for Water, Landscape, and Critical Processes)*
- *Proceedings of the National Academy of Sciences*
- *Nature Communications*

- *Journal of Climate*
- *Geophysical Research Letters*
- *Environmental Research Letters*
- *Journal of Geophysical Research – Atmospheres*
- *Journal of Hydrology: Regional Studies*